

LITHOLOGIC AND PEDOGENIC INFLUENCES ON POROSITY DISTRIBUTION AND GROUNDWATER FLOW IN FRACTURED SEDIMENTARY SAPROLITE: A NEW APPLICATION OF ENVIRONMENTAL SEDIMENTOLOGY

STEVEN G. DRIESE, LARRY D. MCKAY, AND CLINT P. PENFIELD

Department of Geological Sciences, University of Tennessee, Knoxville, Tennessee 37996-1410

e-mail: sdriese@utk.edu

ABSTRACT: Soil and fractured saprolite formed from weathered Middle Cambrian sedimentary rocks were analyzed from two shallow (< 2.5 m deep) pits located at the Oak Ridge Reservation in eastern Tennessee. The effects of both parent heterogeneity and pedogenic processes on fracture and matrix porosity were examined using thin-section petrography and geochemistry. The A and Bw horizons of the soil profiles comprise thin (20–60 cm), weakly developed, loamy inceptisols characterized by high (40–50%) matrix porosity. Subjacent saprolite with pore-occluding pedogenic clay (Crt horizon) occurs beneath a saprolite without pedogenic clay (Cr horizon) and varies lithologically depending on parent material (limestone, siltstone and shale, or sandstone). The soil and saprolite commonly contain root pores; the saprolite additionally contains bedding-parallel and bedding-perpendicular (orthogonal) fracture sets. The fractures and root holes were initially expected to control “bulk” or mesoscale hydraulic conductivity of the materials. This study indicates, however, that extensive pedogenic clay and Fe/Mn-oxide accumulations in the pores and fractures in the saprolite, at depths of 100–250 cm, cause porosity reduction that is so extensive it serves to create a low-hydraulic-conductivity barrier within the upper vadose zone. This barrier is known from prior work to cause formation of perched water-table conditions and rapid downslope flow during rainfall events, but this is the first study to investigate the origin of the low-conductivity barrier.

Geochemical mass balance, using Ti as an immobile index, also demonstrates that three distinct weathering environments defined by element translocation patterns occur in each pit, which correspond to the soil zone (A and Bw horizons), shallow saprolite (Cr horizon), and deep saprolite (Crt horizon). The geochemical zonation is also consistent with the pedogenic and hydraulic interpretation of the system. The potential environmental significance of the zone of pore occlusion is also discussed from the perspective of runoff, recharge, and contaminant migration.

INTRODUCTION

A conceptual model for groundwater flow in fine-grained sedimentary rocks at the Oak Ridge Reservation (ORR) in eastern Tennessee was proposed by Solomon et al. (1992), who defined four broad hydrologic zones, namely the stormflow, vadose, groundwater, and aquiclude zone (Fig. 1). The model is based on studies by numerous researchers including Moore (1988, 1989), Wilson and Luxmoore (1988), and Wilson et al. (1990). Solomon et al. (1992) estimated that 90% of active subsurface flow occurs in the 1–2 m deep stormflow zone, with most of this water flowing rapidly downslope before discharging into seasonal or perennial streams or wetlands. The remaining 10% infiltrates down through the vadose zone and then tends to again move laterally in the saturated zone, which typically begins just above the bedrock-saprolite transition zone, but which is also known to fluctuate seasonally.

The existence of rapid lateral flow in the stormflow zone was confirmed by a field-scale tracer experiment (Wilson et al. 1993) in which a bromide salt tracer released into the stormflow zone was observed to migrate 65 m downslope in only 3.2 h. Peaks in bromide concentration at the downslope monitoring sites occurred during subsequent storms over a one-year mon-

itoring period and were expected to continue for many years. The persistence of bromide in the soil is believed to be due to diffusion into the fine pore structure of the soil and subsequent release during the storm events (Wilson et al. 1993). The potential for rapid migration of solutes in the stormflow zone is of great environmental interest, because large volumes of hazardous wastes, including radionuclides and organic solvents, were buried in numerous shallow pits excavated into the saprolite (Solomon et al. 1992; Shevenell et al. 1994). These pits, although covered with soil caps, are often subjected to seasonal flooding or “bathtubbing,” so contaminants from the lower parts of the pits can be carried up into the stormflow zone, where they can be rapidly transported (Luxmoore and Ferrand 1993). Many of the recent remedial activities at the ORR focus on trying to isolate the waste pits, but because large areas of soil downslope of the pits have already been contaminated, migration through the stormflow zone is expected to be an environmental concern for many years to come.

Although numerous hydrologic, soil, and fracture mapping studies have been carried out in the saprolite and bedrock of the ORR, most of these studies did not directly address the factors causing formation of the stormflow zone. Various researchers, including Dreier et al. (1987) and Solomon et al. (1992), observed numerous fractures and root holes in the upper 1–2 m of the saprolite and hypothesized that these features play a major role in controlling flow. Solomon et al. (1992) stated that measured hydraulic conductivity values in the stormflow zone were as much as 1000 times greater than values in the underlying saprolite, but they did not speculate on possible causes of the observed differences. At the Special Waste Storage Area 7 (SWSA 7) site, hydraulic conductivity values exhibiting a wide range (over four orders of magnitude; Fig. 2) have been measured in the upper 2 m of fractured sedimentary rock saprolite, using both field tests and tests in relatively undisturbed 8–25 cm diameter columns of saprolite in which flow was parallel to bedding. These values are somewhat higher than those measured in sedimentary rock saprolite in the saturated zone at nearby sites, where values typically range from 8×10^{-9} to 4×10^{-5} m/s (Lee et al. 1992; Sanford and Moore 1994; McKay et al. 1997). The authors are not aware of any single site on the ORR where reliable hydraulic conductivity values have been measured in all of the major hydrogeologic zones in the saprolite, namely, the stormflow zone, the vadose zone, and the saprolite portion of the saturated zone.

Investigators working in North Carolina in saprolite formed from crystalline bedrock have also observed zones of lower hydraulic conductivity, often occurring at the interface between the soil zone and the saprolite. In one such study Schoeneberger et al. (1995) measured profiles of saturated hydraulic conductivity in saprolite derived from felsic gneiss and schist, and observed that minimum values tended to occur at depths ranging from 50 to 100 cm. In another study, Vepraskas et al. (1996) and Buol et al. (2000) also observed a zone of very low hydraulic conductivity values below the soil zone and used thin-section microscopy to investigate pore size distribution in this zone. Vepraskas et al. (1996) and Buol et al. (2000) found that most of the macropores (including fractures) were plugged with illuviated clays and other materials, and concluded that the infillings, rather than differences in saprolite texture or structure, were responsible for the lower hydraulic conductivity values. Because of the presence of this zone it is likely that flow conditions similar to those observed in the stormflow zone at the ORR (i.e., perched-water-table conditions and rapid downslope

weathering zones

hydrologic zones

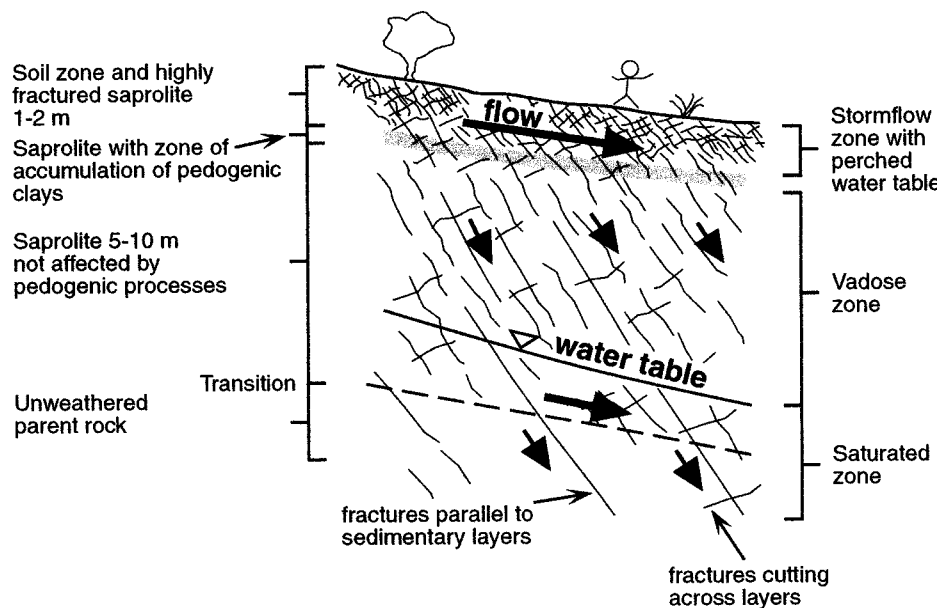


FIG. 1.—Conceptual diagram of conceptual model for flow adapted from Solomon et al. (1992) showing hydrologic zones (stormflow, vadose, and saturated zones) as well as weathering zones, including soil zone (A and Bw horizons) and saprolite (Cr horizon) for Oak Ridge Reservation (ORR), Tennessee. Shaded area shows upper limit of saprolite containing pedogenic clay-filled fracture and matrix pores (Cr horizon) located within 1–2.5 m of the soil surface. Maximum depth of extent of pedogenic influences on saprolite hydrological properties are uncertain.

flow) occur at some sites underlain by crystalline rock saprolite. It is also possible that the cause of the zone of low hydraulic conductivity in both materials (i.e., clogging of macropores with illuviated clays) may be the same.

The overall objectives of this paper are to determine whether clogging of fractures and macropores with illuviated clays and other pedogenic infillings has occurred in the saprolite at SWSA 7 and whether it is sufficiently extensive to be the probable cause of the previously documented reduction in hydraulic conductivity at the base of the stormflow zone at the ORR. Other objectives include determining whether the degree of clogging varies significantly with local-scale variations in lithology of the saprolite and whether the extent of chemical leaching and trace-element translocation are consistent with the conceptual flow model and the hypothesized clogging mechanisms. The paper also includes a discussion of the environmental significance of the zone of pore occlusion and hydraulic conductiv-

ity reduction, and how understanding its origin can aid in environmental investigations and decision-making. In addition, because this study incorporates sedimentologic, pedologic, and hydrologic methods in a multidisciplinary approach, we advocate it as an example of a new application in *environmental sedimentology*. Environmental sedimentology has previously focused mainly on the characterization and three-dimensional simulation of aquifers (e.g., Webb 1994; Neton et al. 1994; McCloskey and Finnemore 1996; Webb and Davis 1998). We believe that sedimentologic approaches can be more widely utilized in the environmental field, particularly in dealing with the very complex physicochemical systems that develop in materials on the borderline between sedimentary rock and soil.

LOCATION AND SETTING

The SWSA 7 site is located in the Valley and Ridge physiographic province at the ORR, 48 km west of Knoxville, Tennessee (Fig. 3). The ap-

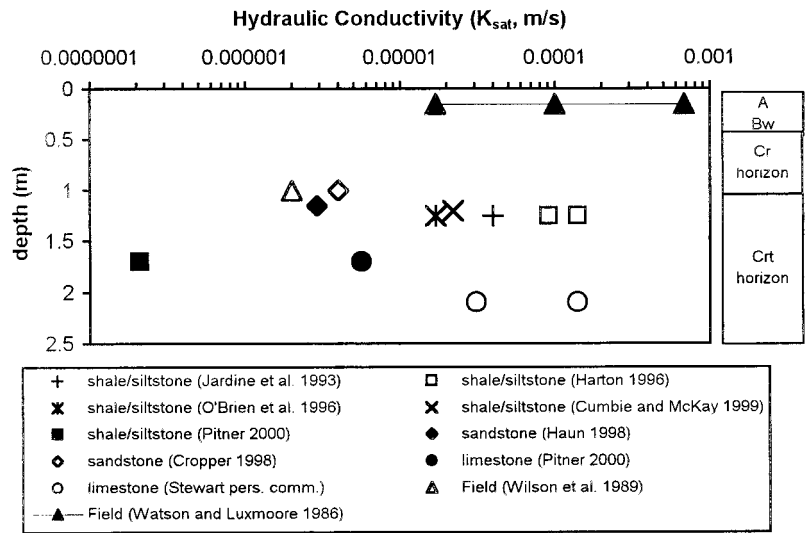


FIG. 2.—Compilation of hydraulic conductivity values (K_{sat} , m/s) versus depth in weathering zone for Special Waste Storage Area 7 (SWSA 7) field site, ORR, noting dominant (>50%) type of saprolite parent material. Laboratory measurements were made using 8.5–25 cm diameter columns of undisturbed saprolite. Measurements for samples recorded as < 1.5 m are from upper pit site, and those from > 1.5 m are from lower pit. The field values plotted for Watson and Luxmoore (1986) represent the geometric mean and range of 30 tests, which had a coefficient of variation of 95%. The field value plotted for Wilson et al. (1989) is the geometric mean of 41 tests, which had a coefficient of variation of 268%. Relative proportions of three principal saprolite lithologies were estimated visually from cross sections of endcaps of dissected saprolite columns. Note also that most K_{sat} measurements were made for weathered material sampled from the Crt horizon, which consists of saprolite with extensive clay pore fillings and coatings (see Table 1).

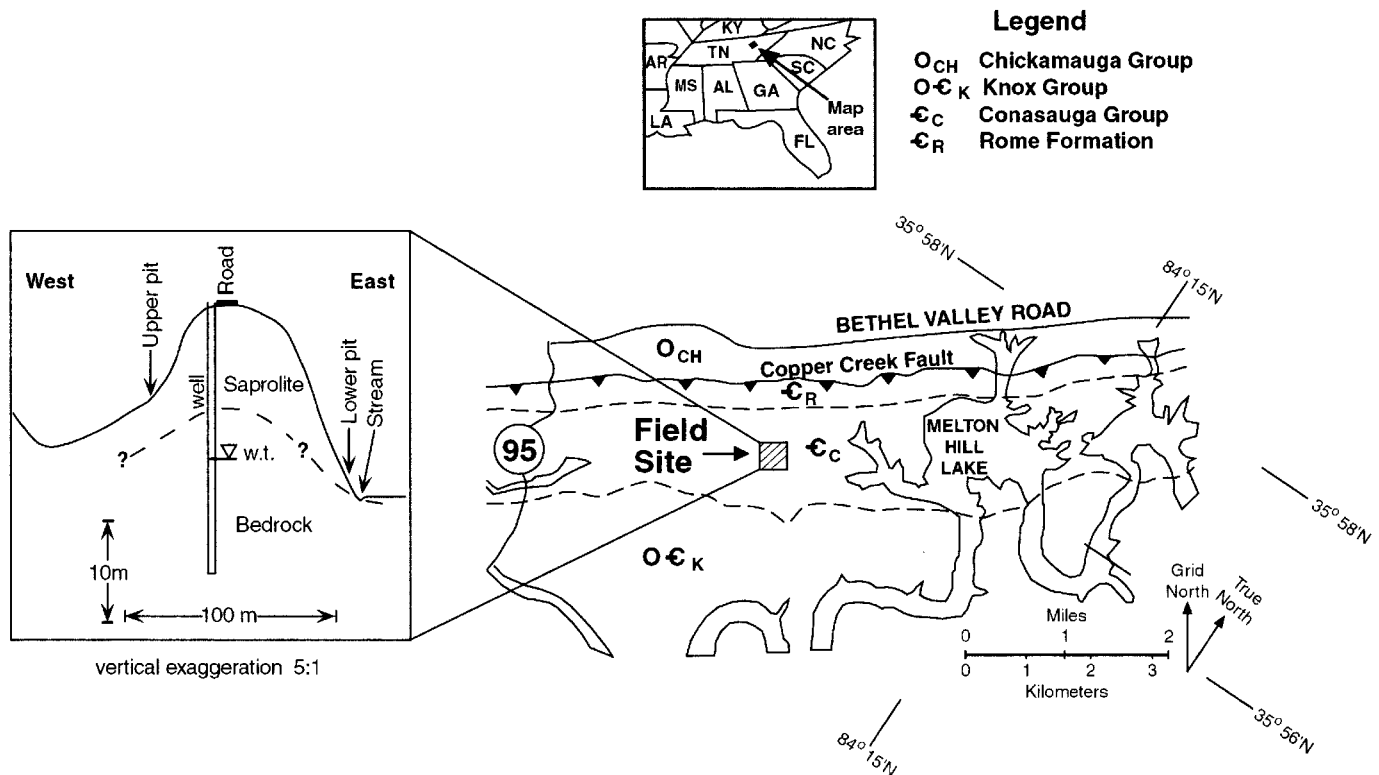


FIG. 3.—Location map showing bedrock geology of ORR and SWSA 7 site in eastern Tennessee, after Hatcher et al. (1992). Inset diagram shows generalized topographic profile and cross section, locations of upper and lower pit study sites, and approximate bedrock-saprolite and vadose zone-saturated zone contacts.

proximate latitude and longitude are 35°55'N, 84°18'W. The climate is humid temperate, and mean annual precipitation was 133 cm during the period 1954–1983. The wettest period lasts from January through March, when the precipitation is 13–16 cm/month. The driest period extends from August through October, when precipitation is 7–10 cm/month. The mean annual air temperature is 15°C (Solomon et al. 1992).

The bedrock at the ORR consists of Lower Cambrian to Lower Silurian carbonates, sandstones, siltstones, and shales. The rocks were deformed by convergent tectonism during the Alleghenian Orogeny (Pennsylvanian to Permian), and crop out in imbricate thrust sheets that strike 030° to 050° and dip 20–35° SE (Hatcher et al. 1992). Complex fracture systems developed in rocks throughout the region because of the initial deformation, and because of subsequent unloading (Dreier et al. 1987; Lemiszki et al. 1992).

DESCRIPTION OF THE SWSA 7 SITE

The SWSA 7 site is underlain by the Dismal Gap Formation (Middle Cambrian), formerly known as the Maryville Limestone (Conasauga Group). The Dismal Gap Fm. is characterized by peloidal and skeletal packstone to grainstone, calcareous very fine-grained sandstone, and siltstone tempestite beds that are locally oolitic and glauconitic (Weber 1988; Foreman et al. 1991; Dreier et al. 1992; Srinivasan and Walker 1993). Limestone beds commonly include flat-pebble conglomerate and range from thin bedded to medium bedded (5–15 cm). Siltstone, mudstone, and shale are interbedded with limestone beds. Siltstone is the volumetrically most abundant of the interbedded lithologies. Siltstone color ranges from gray to gray-green, and it is laminated to thin bedded and locally calcareous.

Two pits were excavated on opposite slopes of a forested ridge (Fig. 3). The 2.5 m deep lower pit is at the toe of the moderately to steeply sloping (40%), northeast-facing side of the ridge (Figs. 3, 4). The 1.5 m deep upper pit is located approximately 100 m southwest along strike at the mid-slope

on the moderately sloping (20%), southwest-facing side of the ridge (Figs. 3, 5). The depth from the base of the upper pit to bedrock is uncertain, but on the basis of data from a nearby well it probably exceeds 5 m (P.M. Jardine, personal communication 1998). The depth from the base of the lower pit to bedrock is <1 m, from our own test borings with a hand auger (Table 1). The lower pit is adjacent to a seasonal stream (Melton Branch Creek) in which the Dismal Gap Fm. crops out.

Soil and saprolite developed from the weathering of Dismal Gap Fm. is generally devoid of calcite or dolomite, leaving illite, smectite, kaolinite, amorphous Fe and Mn oxides, and quartz as the principal soil-forming minerals (Armstrong and Turner 1988; Lee et al. 1990; Jardine et al. 1993; Kooner et al. 1995). Values of pH typically range from 3.5 to 6, and values of cation exchange capacity (CEC) range from 7 to 16 cmol/kg (Jardine et al. 1993). At the SWSA 7 site, soil and saprolite in the upper pit are characterized by a lower pH (3.5–4.5) than that in the lower pit (5–6.5). The soil zone at both sample sites is a thin, loamy-skeletal, mixed, thermic, Typic Dystrachrept to Dystric Eutrochrept (Soil Survey Staff 1990) (Table 1). These are thin, weakly developed soils that have an ochric epipedon and a low base saturation and are generally infertile. The soil zone in each pit has a thin A horizon, which is underlain by a Bw (cambic) horizon (Figs. 4A, B; 5A, B; Table 1). Beneath the Bw horizon in each pit are Cr and Crt horizons (Figs. 4A, C; 5A, C; Table 1). Cr horizon, as used here, denotes saprolite, which is iso-volumetrically weathered rock (in this case, thoroughly leached of carbonate) that still retains primary rock structure such as bedding and fracture sets (Buol et al. 1989; Graham et al. 1994; Stolt and Baker 1994; Cremeens 2000). The Crt horizon designation refers to saprolite that contains illuvial silicate clays as pore linings (Cremeens 2000).

Saprolite lithologies vary according to their parent material (Table 2). Saprolite formed from intraclast (flat-pebble) limestone beds, hereafter referred to as limestone saprolite, has the highest clay content, is very soft and co-

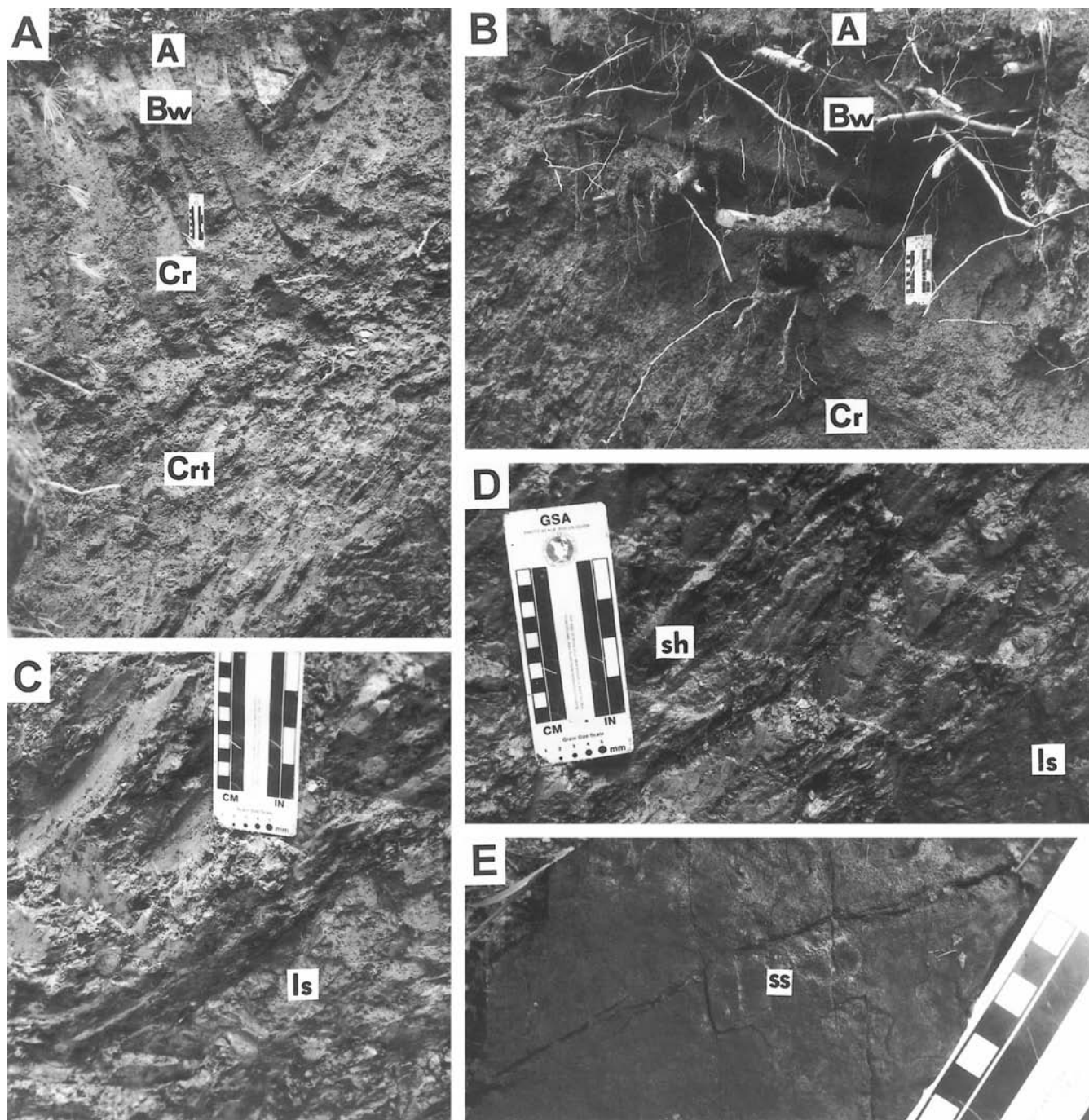


FIG. 4.—Field photographs of soil and saprolite, lower pit sample site, SWSA 7 site, ORR. 15 cm scale card in all photos. **A)** Entire profile showing A, Bw, Cr, and Crt horizons (see Table 1 for complete descriptions). Depth to unweathered bedrock (R horizon) is 323 cm and was established by hand augering. **B)** Densely rooted A and Bw horizons. **C, D)** Bedded saprolite in Crt horizon at depth of 190–230 cm showing limestone (ls) and siltstone/shale (sh) lithologies. **E)** Fractured calcareous sandstone (ss) bedrock (R horizon) exposed in stream bottom near lower pit showing bedding-perpendicular (orthogonal) fractures.

hesive, and is commonly stained reddish-brown to purplish-brown by Fe/Mn oxides (Figs. 4C, 6). A second, and most common saprolite lithology is derived from clayey siltstone interbed; it is hereafter referred to as *siltstone and shale saprolite* (Figs. 4C, 5C, 7). Siltstone and shale saprolite is hard and blocky, and Fe/Mn-oxide staining is pervasive. Beds derived from the weathering of calcareous, very fine-grained sandstone parent material are relatively soft to moderately hard and are moderately cohesive. Hereafter, this lithology is referred to as *sandstone saprolite* (Fig. 6). More detailed descriptions of saprolite are provided below and in Table 2.

ANALYTICAL METHODOLOGY

Physical characterization of the soil and saprolite included observation and description of macroscopic features seen in the field (Table 1), as well as features observable in polished hand specimens and in thin sections (Table 2). Large oriented samples (20–30 cm diameter or larger) were excavated from the soil zone and saprolite of each pit. Cross-sectional slices of the same saprolite were also examined from columns used in previous hydrogeologic experiments (Figs. 2, 6). Unweathered Dismal Gap Fm. bed-

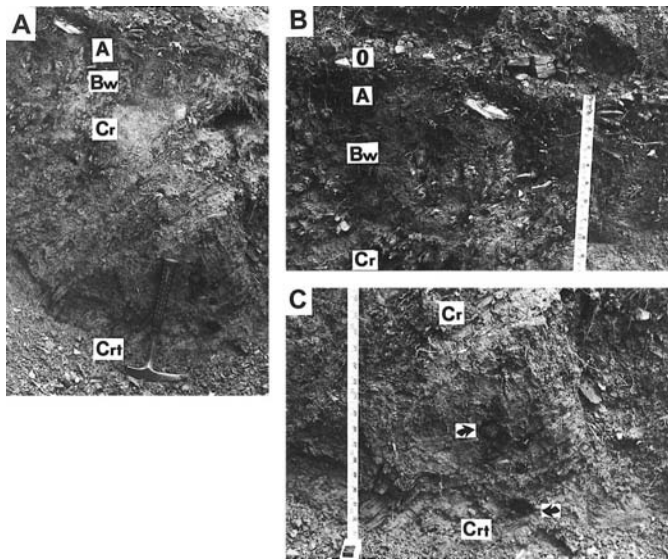


FIG. 5.—Field photographs of soil and saprolite, upper pit sample site, SWSA 7 site, ORR. **A**) Entire profile showing thin (upper 20 cm) heavily rooted soil zone (O, A and Bw horizons), and bedded sedimentary saprolite (Cr and Crt horizons) to depth of 110 cm (see Table 1 for complete descriptions). Hammer for scale. **B**) Densely rooted, thin O, A, and Bw horizons grading into Cr horizon. 25 cm rule for scale. **C**) Cr and Crt horizons with large partially decayed roots (arrows) penetrating the saprolite. 35 cm rule for scale.

rock, including siltstone/shale, limestone, and sandstone, was sampled with a rock hammer from a stream bed that is adjacent to the lower pit (Fig. 4D). Twenty-two soil, saprolite, and bedrock samples were brought back to the laboratory with minimal disturbance. After air-drying and impregnation with polyester boat resin and thin-section epoxy, soil and saprolite thin sections were ground, without the use of a lubricant, to optical thickness (30 μm) using progressively finer grades of silicon carbide sandpaper to 600 grade, as described by Mora et al. (1993). Hand specimens were polished using the same procedure. Bedrock samples were prepared into thin sections by standard petrographic methods. Textures, percent mineralogical and pedogenic constituents, and percent porosity were estimated from thin-section surfaces using standard visual comparison charts (Terry and Chilingar 1955).

Fifty-three 50–100 g bulk samples were collected at 10 cm intervals along vertical profiles through each pit (Appendix 1). Samples were analyzed with X-ray fluorescence (XRF) using an energy-dispersive EG & G ORTEC TEFA III tube-excited Analyzer (Singer and Janitzky 1986). ORTEC's ATAC program and a clay soil and shale protocol were used to analyze selected elements. Bulk density (g/cm^3) was determined by paraffin coating of air-dried soil and saprolite clods (Blake and Hartge 1986). Geochemical interpretation utilized mass-balance reconstruction from calculations involving whole-rock element abundance and bulk density (e.g., Penfield 1999; Driese et al. 1992; Driese et al. 2000; Ashley and Driese 2000).

LITHOLOGIC AND PETROGRAPHIC CHARACTERISTICS

Mesoscale and Microscale Features

Description.—Textural and mineralogical characteristics of the soil and saprolite are summarized in Table 2. Epoxied and polished hand specimens provided initial insight into subtle lithologic heterogeneity. Figure 7A illustrates the boundary between the soil zone (Bw horizon) and the underlying siltstone and shale saprolite (Cr horizon). Fragmented saprolite clasts (*lithorelicts*, in soil-science terminology) indicate that colluvial mixing of saprolite into soil matrix has occurred. Saprolite clasts account for a significant proportion of the soil zone in each pit (Table 1). Hand specimens

TABLE 1.—Soil and saprolite descriptions at Special Waste Storage Area 7 (SWSA 7) field site, Oak Ridge Reservation (ORR), Tennessee.

Site position: sideslope (backslope); Slope: 20% (upper site) to 41% (lower site); Parent material: colluvium over residuum, Dismal Gap Fm. (Middle Cambrian) interbedded calcareous siltstone, shale, limestone and sandstone; Drainage: No standing water detected; Rooting: Roots down to the pit bottom; Vegetation: mixed hardwoods

Classification: loamy skeletal, mixed, thermic, shallow Typic Dystrachrept or Dystric Eutrochrept

Horizon	Depth (cm)	Description
Site (1) upper pit: dissected upland shoulder position a meter below a narrow rounded summit.		
Oi	+3 to +1	Leaf litter, twigs, bark.
Oa	+1 to 0	Unrecognizable (degraded) organic matter.
A	0 to 8	Very dark grayish-brown (10YR 3/2) shaly clay loam; moderate fine granular structure; very friable; common fine roots; 35–50% siltstone fragments; pH 5.0; clear wavy boundary.
Bw	8 to 18	Yellowish-brown (10YR 5/4) very shaly loam; weak fine subangular blocky structure with pockets of rock structure; very friable; common fine roots; 50–75% siltstone fragments; pH 3.9; clear wavy boundary.
Cr	18 to 95	Light olive brown (2.5Y 5/4) soft, highly fractured, non-calcareous (leached) siltstone; massive; complex micro-folded rock structure; all rock fragments coated with yellowish-brown (10 YR 5/4) soil material; loose after being disturbed; 90–95% siltstone fragments 1 to 3 cm in mean length; few fine roots; pH 4.2; clear wavy boundary.
Crt	95 to 110+	Light olive brown (2.5Y 5/4), highly fractured, non-calcareous (leached) siltstone, dense in place but loose after disturbance; most rock fragments and fracture surfaces coated with common moderate yellowish-brown (10 YR 5/4) ferrans and common dusky brown (5YR 2/2) mangans below a greenish-gray (5 GY 6/1) to light olive gray (5Y 6/1) clay-plugged zone that occurs at the boundary between Cr and Crt; clay occurs as common coatings on root pores, fracture surfaces, and in intergranular pores; very few fine roots; pH 4.0; estimated depth to R horizon is 5 to 10+ meters, based on test boreholes.
Site (2) lower pit: toe of slope near Melton Branch Creek.		
Oi	+5 to +2	Leaf litter, twigs, bark.
Oa	+3 to +1	Unrecognizable (degraded) organic matter.
A	0 to 15	Light olive brown (2.5Y 5/3) loam; common fine distinct yellowish-brown (10YR 5/8) mottles; weak fine subangular blocky structure; very friable; many fine and medium roots; 3% colluvial rock fragments (rounded corners); abrupt smooth boundary.
Bw	15 to 60	Pale yellow (2.5Y 7/4) loam to gravelly loam; common fine distinct yellowish-brown (10YR 5/8) mottles; weak fine subangular blocky structure; friable; many fine roots; 20% sandstone and siltstone fragments; some fragments are colluvial; inside rock fragment color is dark brown; clear smooth boundary.
Cr	60 to 125	Light olive brown (2.5Y 5/4) soft, highly fractured, non-calcareous (leached) siltstone; massive; complex micro-folded rock structure; all rock fragments coated with yellowish-brown (10YR 5/4) soil material; loose after being disturbed; 90–95% siltstone fragments 1 to 3 cm in mean length; few fine roots; clear wavy boundary.
Crt1	125 to 225	Light olive brown (2.5Y 5/4), highly fractured, non-calcareous (leached) bedded siltstone, dense in place but loose after disturbance; most fracture surfaces coated with few moderate yellowish-brown (10YR 5/4) ferrans and common dusky brown (5YR 2/2) mangans below a greenish-gray (5GY 6/1) to light olive gray (5Y 6/1) clay-plugged zone that occurs at the boundary between Cr and Crt1; clay occurs as common coatings on root pores, fracture surfaces, and in intergranular pores; very few fine roots; (base of pit exposure)
Crt2	225 to 323	Light olive brown (2.5Y 5/4), highly fractured, non-calcareous (leached) siltstone; dense, variable hard to soft; most rock fragments and fracture surfaces coated with common moderate yellowish-brown (10YR 5/4) ferrans and common dusky brown (5YR 2/2) mangans; greenish-gray (5GY 6/1) to light olive gray (5Y 6/1) clay occurs as common coatings on root pores, fracture surfaces, and in intergranular pores; no roots; (auger sample description)
R	323+	Hard, dense, weakly calcareous siltstone; refusal of auger

of the limestone saprolite consist of *in situ* weathered intraclasts (i.e., flat-pebble limestone conglomerate clasts) originally present within the limestone parent (Fig. 4B). Irregularly distributed clay pore fillings are common and vary in color from yellowish brown to greenish-gray (Fig. 7B). Porosity, with some pore apertures approaching 2 mm in diameter, occurs between intraclasts where pore-filling clay is absent or less abundant, and also in unfilled root voids. Siltstone and shale saprolite hand specimens exhibit the highest density of fractures, expressed as a fracture spacing of 0.5–3 cm (Fig. 7C). Bedding-parallel fractures extend the entire length of hand specimens and are interconnected with bedding-perpendicular frac-

TABLE 2.—Summary of petrographic and geochemical properties of soil zone and saprolite sampled from SWSA 7, ORR, Tennessee.

Geologic/Pedologic Material	Detrital Mineralogy	Pedogenic Features	Matrix Porosity	Fracture Porosity
Soil Zone (A and Bw horizons); thin (20–60 cm), loamy-skeletal, mixed, thermic, Typic Dystrachrept (Soil Survey Staff 1990)	40–60% monocrystalline quartz; 2% glauconite; 2–5% K-feldspar; <1–2% plagioclase; <2% muscovite + biotite; <1% carbonate allochems; <2% heavy minerals	25–40% lithorelicts (saprolite clasts); 2–6% Fe/Mn-oxide/oxyhydroxide glauconites; 1–20% very coarse to fine roots; <1% pedogenic clay	Very high (40–50%), mostly intergranular and in root channels; decreases progressively downward to contact with saprolite	None; no soil fractures are apparent in field or in thin section
Sandstone saprolite derived from weathering of fractured, very fine-grained, parallel- to ripple cross-laminated, glauconitic, peloidal, feldspathic quartz arenite; estimated 20% of saprolite section	55–65% monocrystalline quartz; 5–8% glauconite; 5–10% K-feldspar; <5% plagioclase; 2–5% muscovite + biotite; <2% carbonate allochems; <2% heavy minerals	vadose pedogenic clay cements at detrital grain contacts (meniscus) and on undersides of grains (pendant); pedogenic clay infillings of fractures; Fe/Mn oxide and oxyhydroxide coatings on grain surfaces and in fractures; <1% roots	High (20–30%), mostly in intergranular pores and to a lesser degree in dissolution intragranular (grain moldic) pores; matrix pore apertures range from 1–100 μm , with modal size in the 20–50 μm size range	Few fracture pores (<5%) unfilled with pedogenic clays, with average spacing of 0.5–5 cm; fracture apertures (pre-clay infill) range from 5.0 μm to 0.5 mm, but open fractures range from 1–10 μm
Siltstone/shale saprolite derived from weathering of fractured, laminated to weakly bioturbated, illitic to chloritic clayey siltstone to silty shale; estimated 50% of saprolite section	matrix consists of 50–80% illite, with subordinate chlorite; 10–30% monocrystalline quartz; 5–10% detrital biotite and muscovite; <1% glauconite; <1% heavy minerals	pedogenic clay and Fe/Mn oxide and oxyhydroxide infillings and coatings of fractures; <2% roots	Probably high (30–50%), but cannot be resolved with petrographic microscope; matrix pore apertures probably <1 μm	Common fracture pores (10–20%), with average spacing of 0.5–3 cm; fracture apertures (pre-clay infill) range from 5.0 μm to 1.5 mm, with modal size of 10–20 μm for open fracture pores
Limestone saprolite derived from weathering of fractured, intraclastic rudstone to floatstone that consists of 0.5–10 cm diameter, discoidal intraclasts comprised of sandy, peloidal and skeletal lime grainstone to packstone; estimated 30% of saprolite section	matrix consists of mixture of 80–90% illite and random (disordered) smectite/ chlorite; 10–40% monocrystalline quartz; 2–5% detrital biotite and muscovite; 95% glauconite; <1% heavy minerals	abundant pedogenic clay random (disordered) smectite/chlorite or smectite/vermiculite; pedogenic clay and Fe/Mn oxide and oxyhydroxide pore-fillings and coatings of fractures; 2–5% roots	High (30–50%), includes large root pores (up to 5 mm) in matrix, but many other matrix pores cannot be resolved with petrographic microscope; most matrix pore apertures probably <5 μm	Few fracture pores (<5%) unfilled with pedogenic clays, with average spacing of 0.5–5 cm; fracture apertures (pre-clay infill) range from 5.0 μm to 2 mm

(Estimated using 4–5 thin sections per lithology and Terry and Chilingar (1955) visual comparison charts.)

tures, which commonly occur as steeply inclined (45–90°) conjugate orthogonal pairs that do not always extend across beds. Fractures in hand specimens of the sandstone saprolite are also closely spaced (0.5–5 cm), completely crosscut individual centimeter-thick beds, and are mostly of the bedding-perpendicular type. The widest fractures crosscut multiple beds of different lithologies (Fig. 7D). These fractures are typically filled with yellowish-brown to greenish-gray clay. Some porosity also resulted from shrinkage during sample preparation.

Petrographic features, including detrital mineralogy, texture, pedogenic features, and mineralogy of the soil zone and saprolite are summarized in Table 2. Samples from the Bw horizon consist of a loamy, silty matrix that contains 2–6% Fe/Mn-oxide glauconites and 25–95% Fe/Mn-stained colluvial saprolite fragments, as well as reworked pedogenic clay coatings (Fig. 8A, B). Samples from the Crt horizon saprolite commonly have fracture,

root, and matrix pores extensively lined, and in some instances filled by, pedogenic clay that has a high birefringence under crossed polarized light (Fig. 8C–G). Weathered glauconite grains are closely associated with both pedogenic clay and Fe/Mn-oxide accumulations (Fig. 8F, G). Bedrock samples, in contrast, contain abundant calcite spar cement in matrix and fracture pores but do not contain birefringent clay or Fe/Mn-oxide accumulations (Fig. 8H).

Microscale fracture porosity is summarized in Table 2. A quantification of saprolite porosity observable in thin section was not attempted because of “artificial” porosity created by sample drying during thin-section preparation. Nevertheless, rough visual estimates of total porosity (matrix + fracture) were made using standard visual comparator charts (Table 2). Saprolite macropores exhibit irregular apertures ranging from 5 to 250 μm and have a mode in the 20–50 μm range (Fig. 8C–H). The fracture system

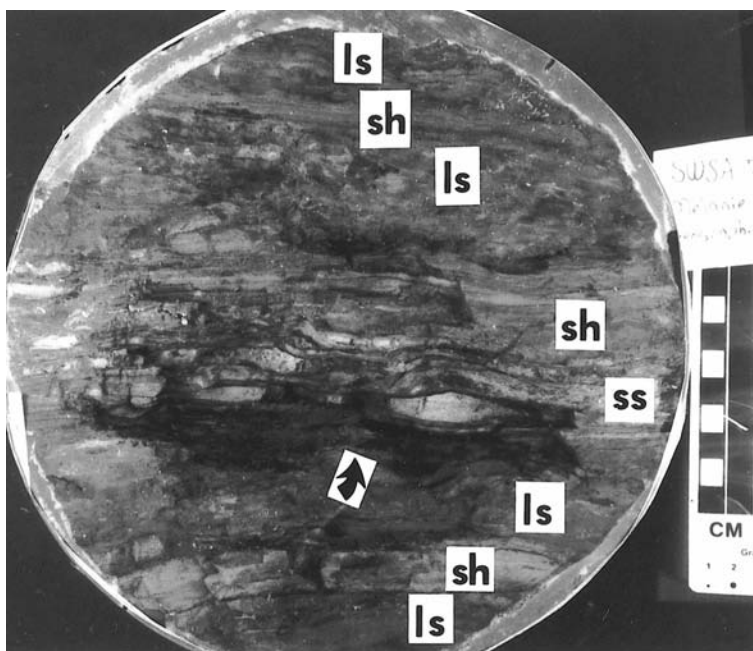


FIG. 6.—Cross section of 25 cm diameter, dissected experimental saprolite column from lower pit site (M. Stewart, personal communication 2000), sampled from depth of 1.9–2.3 m (Crt horizon), SWSA 7 site, ORR. Note complex, fine-scale interbedding of three distinct lithologies including clay-rich limestone (ls), siltstone/shale (sh), and sandstone (ss) saprolites, as well as extensive Fe/Mn-oxide staining of fracture surfaces (arrow, darker-colored areas).

in the siltstone saprolite appears most uniform at the thin-section scale. The dominant orientation consists of long, bedding-parallel fractures with bedding-perpendicular fractures that extend from bedding-parallel fractures. Some samples exhibit complex anastomosing fractures. Bedding-perpendicular fractures are more laterally extensive, whereas bedding-parallel fractures rarely terminate without extending across the entire bed or intersecting another fracture. Observed fracture pore apertures, including infilling (if present), range from less than 10 μm to 1.5 (+) mm, with the mean aperture of the siltstone/shale saprolite (10–20 μm) less than that observed in the sandstone saprolite (20–50 μm), and with bedding-perpendicular fractures constituting the larger fractures. These observations are compatible with measurements made by Dorsch and Katsube (1998, 1999) using helium and mercury porosimetry, in which geometric mean pore apertures of 100–200 μm were calculated for the sandstone saprolite versus 10–40 μm for the siltstone saprolite. The sample of limestone saprolite sampled by Dorsch and Katsube (1998) had an exceptionally high geometric mean pore aperture of 1170 μm , which is compatible with our own observations of very large but highly irregular root pores up to 5 mm in diameter in this lithology.

Interpretation.—Complete leaching of carbonate cement and allochem grains, after weathering (i.e., “saprolitization”) and formation of a Cr horizon, could have resulted in an increase in saprolite matrix and fracture porosity. However, any enhanced matrix and fracture porosity in the saprolite was subsequently pedogenically modified by clay and Fe/Mn oxide cements, which fill intergranular matrix and fracture pores. Pedogenic processes produced diffuse Fe/Mn-oxide coatings on the outer margins of detrital grains, as well as Fe/Mn cementation and matrix impregnation (Fig. 8C–F). Pervasive pedogenic clay effectively occludes the overwhelming majority of the larger matrix pores in both the sandstone and limestone saprolites, forming vadose pendants on the undersides of pores, and coatings that line root pores and fracture pores, some of which exhibit complex microbanding, which attests to multiple generations of illuviation (Fig. 8C–F; FitzPatrick 1993). The types of coatings include silt, MnO_2 -clay compound, and composite (classification of FitzPatrick 1993).

The discrepancy between the saprolite porosity we determined using the wax clod method (42–55%; Appendix 1) and that determined using Hg (14–51%) and He (18–53%) porosimetry by Dorsch and Katsube (1998, 1999) may be because their porosimetry measurements were made on smaller saprolite aggregates than our wax clods, and hence contained fewer larger natural pores. Also, the Hg and He porosimetry measurements cannot easily be carried out on weak or friable samples, hence there is a tendency to bias the samples by collecting more competent clods, which may have a lower porosity. The fracture systems in the saprolite appear more complex in thin section than are apparent at the hand-specimen scale. Not only are there more observable fine fractures, but the fractures exhibit a higher amount of apparent “interconnectedness,” with conjugate bedding-perpendicular pairs that intersect bedding-parallel fractures (Fig. 8C, D). Whereas in polished hand specimens the fractures in the saprolite appear to have measurable pore aperture, many fractures observed at the microscopic level are so extensively occluded with pedogenic clay and Fe/Mn-oxides that there appears to be little or no open fracture porosity (Fig. 8C–G).

GEOCHEMISTRY OF SOILS AND SAPROLITES

Mass-Balance Reconstruction

Description.—The behavior of chemical constituents in both the soil and the saprolite was evaluated using a mass-balance approach, following Brimhall et al. (1988 Brimhall et al. 1991a, Brimhall et al. 1991b). The approach is generally used to evaluate chemical variations due to both the closed-system effects of residual enrichment and volumetric changes in the soil matrix, and the open-system transport of material into or out of the soil (Driese et al. 1992; Driese et al. 2000; Ashley and Driese 2000). The

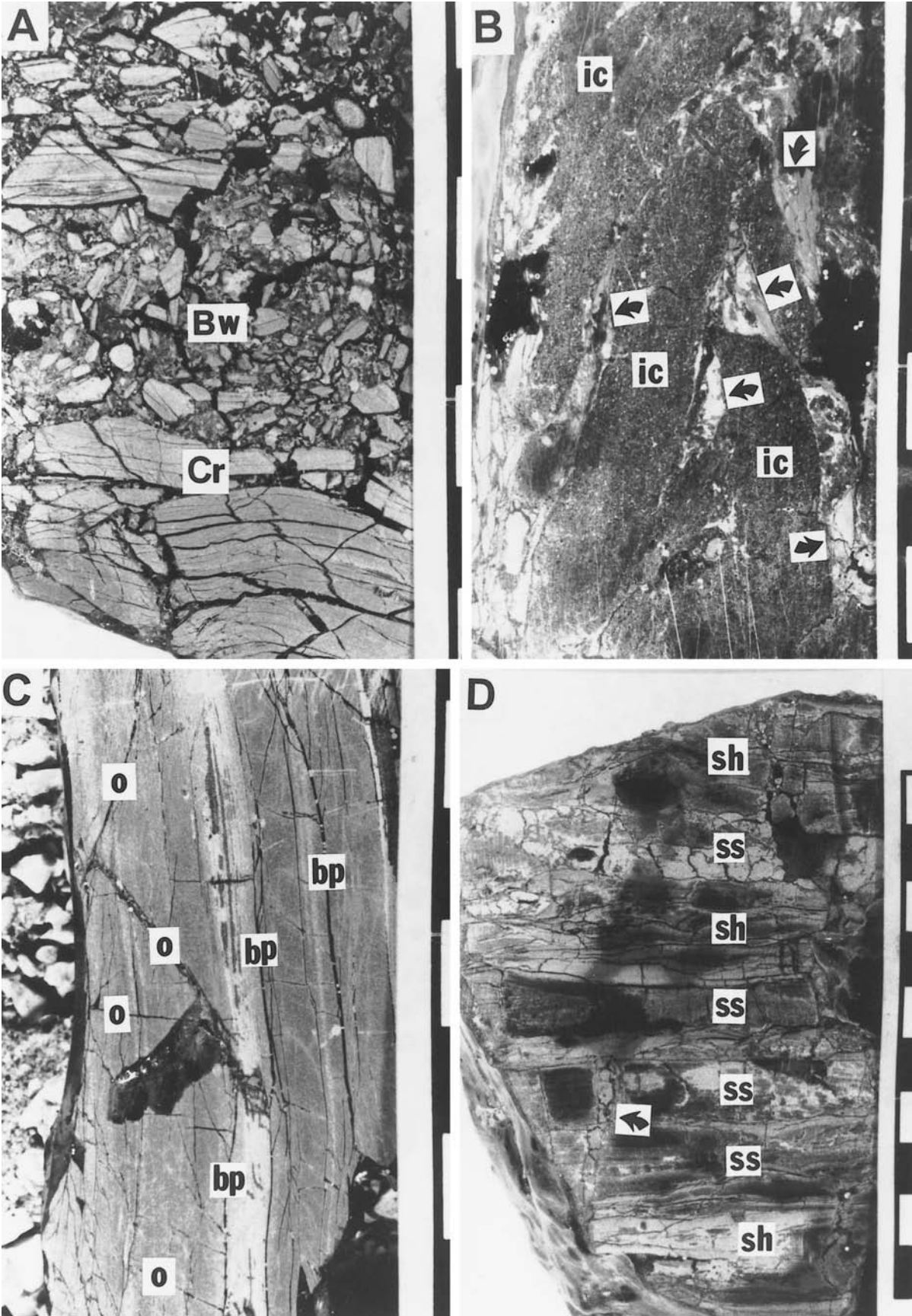
calculations for the closed-system effects are necessary to obtain values for the open-system transport function.

The open-system transport function, τ , evaluates the net gain ($\tau > 0$) or net loss ($\tau < 0$, to a minimum of -1 , which equates to 100% loss) of a chemical constituent relative to the representative parent and with respect to an immobile index element, typically Zr or Ti (Gardner 2000; Stolt and Baker 2000). In order to quantify the more subtle chemical modifications of the saprolite due to depth-related changes in pedogenic clay and Fe/Mn-oxide concentrations, a lowest, least-weathered, carbonate-free saprolite sample was selected, instead of unweathered calcareous bedrock, as the most reasonable model parent material. The bulk composition of this deepest saprolite (250 cm depth, lower pit) permitted geochemical comparisons between samples and parent to be made on a “carbonate-free” basis (cf. Stolt and Baker 2000). The immobile index element selected for the mass-balance calculations was Ti, because it produced the most conservative calculations for volumetric changes during weathering and had a more consistent concentration with depth in each pit (cf. Gardner 2000). The chemical behavior of Na, K, Mg, and Ca changes across the transition from the soil zone (Bw horizon) to the shallow saprolite zone (Cr horizon; below 30 cm in upper pit and below 70 cm in the lower pit) (Fig. 9A, B). All elements experience net gain below the transition, except for Na, which is lost throughout the shallow saprolite zone and is gained in the deep saprolite zone (Crt horizon, below 100 cm in the upper pit and below 140 cm in the lower pit); greater changes in chemical behavior across this zone occur in the lower pit. Na exhibits the greatest fluctuations throughout each profile, and the greatest enrichments occur in the deep saprolite zone (Crt horizon), where pedogenic clay accumulation is greatest, as observed petrographically (Figs. 7, 8). K also varies inversely with Na throughout the saprolite of each profile.

The overall trends of Si and Al indicate that the two elements vary inversely in the soil zone (A and Bw horizons) and shallow saprolite (Cr horizon) of each profile (Fig. 9C, D). Si becomes more depleted with increasing depth into the soil zone, whereas Al becomes less depleted. Si exhibits modest gains in the soil zone and more loss at the transition from soil to shallow saprolite (Bw to Cr horizon; below 30 cm in upper pit; below 60 cm in lower pit), whereas Al experiences a relative gain in each profile, so much so in the upper pit that Al has a net gain (positive value). Rather conservative trends continue for each until the approximate crossover depth into deep saprolite (Crt horizon); beyond this depth in the upper pit, both elements exhibit net gain, and Al does the same in the lower pit. Ba and Sr are inversely related, and the Ba/Sr ratio is highest in the soil zone (A and Bw horizons) of each site, whereas they are positively correlated and the ratio is lowest in both saprolite zones (Cr and Crt horizons) (Fig. 9C, D).

The redox-sensitive elements, such as Mn, Fe, Ni, and Cu (also Co, which is not plotted) all show depth variations that define three distinct zones (Fig. 9E, F). All of these elements are depleted in the soil zone (A and Bw horizons), and, with the exception of Cu, exhibit general depletions in the shallow saprolite zone (Cr horizon) as well. Mn, Fe, Ni, and Cu enrichments are greatest in the deep saprolite zone (Crt horizon) (Fig. 9E, F). V, Cr, As, and Zr show covariant behavior, and the depth functions for these elements also define three zones (Fig. 9G, H). In contrast to some other redox-sensitive elements described previously, these trace elements have net gains in the soil zone (A and Bw horizons) relative to the saprolite (P does also but is not plotted here).

Interpretation.—The behaviors of the exchangeable bases Na, K, Mg, and Ca are mainly functions of position in the profile, clay content, and extent of carbonate leaching (Fig. 9A, B). That K also varies inversely with Na throughout the saprolite of each profile may be the result of cation substitution into different clay lattices (i.e., relative abundance of illite inherited from parent material containing K^+ versus pedogenic smectite/vermiculite containing Na^+ ; see discussion that follows). The net gain of Al in the deep saprolite (Crt horizon) is attributed to its preference for clay



lattices and the observed abundances of pedogenic clay in this interval (Fig. 9C, D). Leaching is also commonly measured by calculating the ratio of wt% Ba to wt% Sr (Retallack 1990). Ba and Sr are inversely related, and the ratio is highest in the soil zone (A and Bw horizons) of each site, indicating higher leaching and weathering (as would be expected). Ba and Sr are positively correlated and Ba/Sr is lower in both saprolite zones (Cr and Crt horizons) of each site, where leaching is reduced and clay content is higher. Enrichments in Mn, Fe, Ni, and Cu (Fig. 9E, F) are greatest in the deep saprolite zone (Crt horizon) and probably record variable Eh conditions (redoximorphy) caused by periodic saturation (reduction) and aeration (oxidation) (Vepraskas 1992; Vepraskas et al. 1994), as evidenced by Fe/Mn pore fillings observed as both macroscopic stains (Fig. 6) and petrographically as pore fillings (Figs. 7, 8). Mn translocations are greater than other redox-sensitive elements because of the ability of Mn to exist in a wide range of mineral phases. Net losses of Mn and Fe in the profiles through the shallow saprolite (Cr horizon) suggest lower Eh in this region (cf. Driese et al. 2000). Net gains for V, Cr, As, and P (and Ca; Fig. 9A, B, G, H) in the soil zone (A and Bw horizons) are attributed to biocycling of these elements by plants. Net gains in Zr, which largely resides in detrital zircon silt grains, likely represents residual enrichment during soil formation (Fig. 9G, H); similar increases in detrital quartz silt and sand in the soil zone due to residual enrichment are reflected by the similar, but weaker, gains in Si in the soil (Fig. 9C, D). Low Eh in the shallow saprolite (Cr horizon) is suggested by depleted V, Cr, and As, especially in the upper pit (Fig. 9G, H). Cr, like Mn, shows the greatest apparent fluctuations, perhaps because it can exist in a wide variety of mineral phases and in different oxidation states. Other redox-sensitive trace elements exhibit similar behaviors characterized by covariation.

DISCUSSION

Petrography and Hydrology of Soils and Saprolites

Our petrographic analysis indicates that the extensive clay and Fe/Mn-oxide accumulations that occur in the saprolite, as both matrix and fracture pore fillings (Figs. 7, 8), are almost certainly a major control on hydraulic conductivity and the downward flow of soil water. The pedogenic clay coatings and pore fillings are best developed at a depth below 105 cm in the upper pit and below 140 cm depth in the lower pit, and this should result in a decrease in hydraulic conductivity at these depths. The degree of pore occlusion at depths greater than 2.5 m is uncertain, primarily because there has been no excavation and sampling of the “deep” saprolite. The variation in K_{sat} values measured in these materials, in both experimental columns and in the field (Fig. 2) must reflect, in part, local spatial variability in the degree of pore plugging by clay and Fe/Mn oxide coatings and pore fillings. These coatings and pore fillings are also expected to be a major factor in causing the frequent perched-water-table conditions observed in previous hydrogeologic studies (Fig. 1). Decreases in fracture and biopore density also occur with depth, but these changes are gradual and are unlikely to be solely responsible for the large reduction in hydraulic conductivity at the base of the stormflow zone.

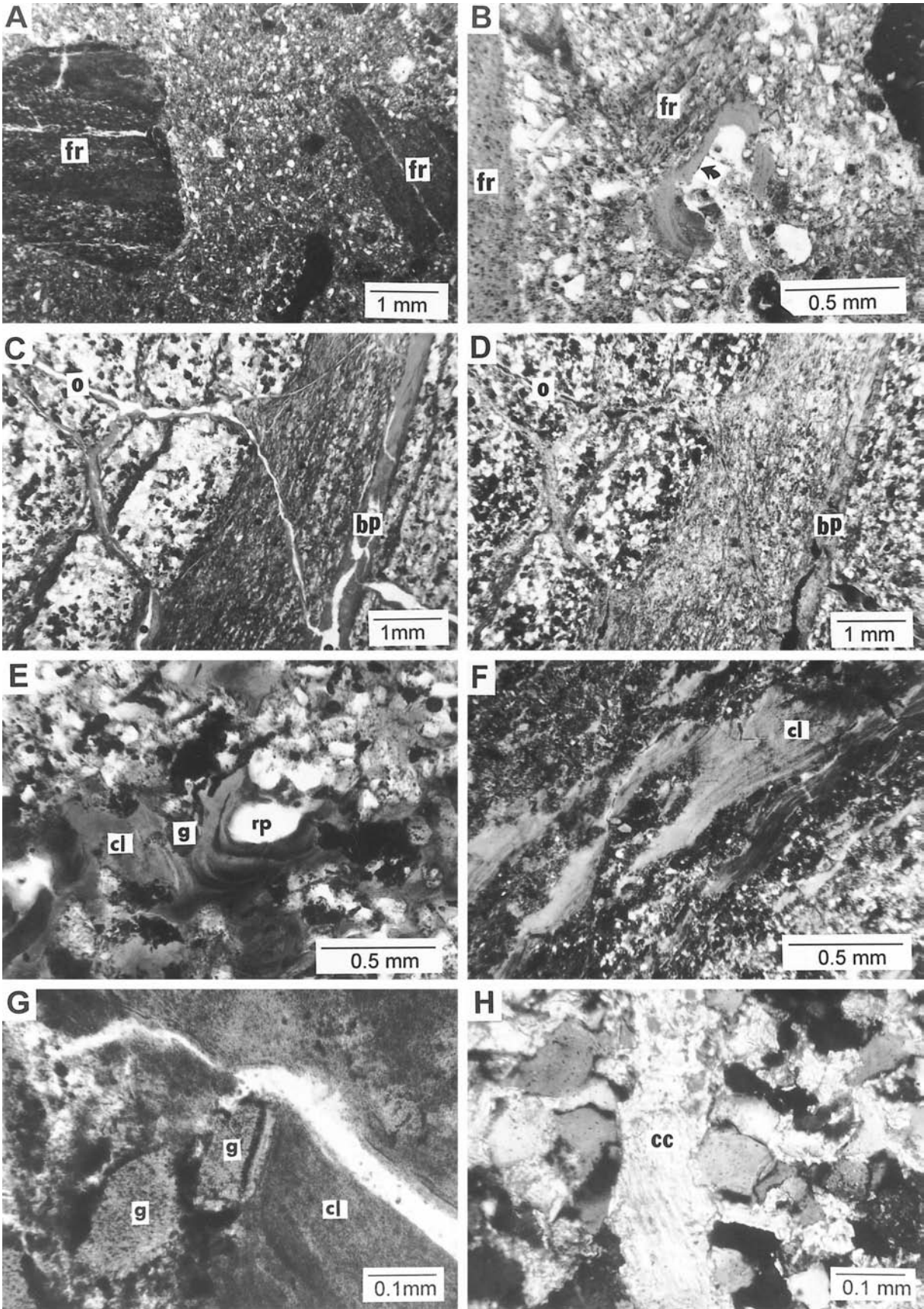
It is not possible, with the present data (Fig. 2; Tables 1, 2; and previously cited papers), to systematically assess the relative influence of the many pedogenic and lithologic factors that are expected to influence hy-

draulic conductivity in the saprolite and soil at the SWSA 7 site. It is possible, however, to make some general comments based on the data, and to pose hypotheses for future studies. First, it is clear that fractures, root holes, and other macropores provide the main control on groundwater flow during saturated conditions, and hence it is reasonable to expect that the degree of infilling of such features is an important factor in controlling flow. Previous field studies by Watson and Luxmoore (1986) indicated that macropores (note: they did not discriminate between different types of macropores or lithologies) dominated flow in at least the upper part (15 cm depth) of the soil and saprolite at the SWSA 7 site. They measured infiltration rates at 30 locations using a tension infiltrometer at tensions of 0, 3, 6, and 15 cm heads of water and observed a very rapid decline in infiltration rate with increasing tension. They calculated that at a depth of 15 cm, 96% of the flow occurred in macropores of 100 μ m diameter or larger and that 82% of the flow occurred in pores larger than 250 μ m in diameter. These macropores are much larger than the typical values for the intergranular matrix (this study; Dorsch and Katsube 1998, 1999), so it confirms that intergranular porosity plays a secondary role in controlling flow.

Additional evidence for active flow in fractures and root holes in all of the three saprolite lithologies was provided by previous experiments on groundwater flow and contaminant transport carried out in six undisturbed saprolite columns excavated at the SWSA 7 site (Harton 1996; Cumbie 1997; Cumbie and McKay 1999; Haun 1998; Pitner 2000; Driese and McKay, unpublished data). These samples were all collected between 1 and 2.3 m depth at the upper and lower pit sites. Values of hydraulic aperture calculated using the cubic law (Snow 1968) ranged from 19 to 110 μ m. These values are smaller than the macropore diameters of up to 250 μ m calculated by Watson and Luxmoore (1986), but the columns were collected mainly below the root zone, so this is to be expected. In several of the experiments, water containing fluorescent latex microspheres (0.05–4.2 μ m diameter) was pumped through undisturbed saprolite columns in order to study retention of colloids from flowing groundwater. Afterwards each of the columns was dissected and the distribution of fluorescence was observed under UV light. In a siltstone and shale saprolite column Cumbie and McKay (1999) observed fluorescence on nearly all of the bedding-perpendicular fractures and approximately half of the bedding-parallel fractures. On a given fracture surface the fluorescence tended to occur in both irregularly shaped regions and in “channels,” with only about 10–30% of the fracture surface having visible fluorescence. Similar results were observed in a column containing 50% sandstone saprolite (Haun 1998; Driese and McKay, unpublished data), again indicating that even though fractures and/or root holes were largely occluded with pedogenic clays and oxides, some portions of these features remained open and were still important pathways for flow. Subsequent preliminary examination of the thin sections from these columns (McKay and Driese 1999) confirm that almost all of the microspheres were retained along fracture or root holes, with very few found in the intergranular pore regions. In a recent study immiscible trichloroethylene (TCE) containing fluorescent dye was injected into two saprolite columns consisting of approximately 50% siltstone/shale saprolite and 50% clay-rich limestone saprolite (Pitner 2000). When the columns were dismantled and examined under UV light, fluorescence was found to occur mainly along fractures in the siltstone and shale saprolite, but in the

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FIG. 7.—Photographs of polished hand specimens showing macroscale and mesoscale features of the solum and saprolite at SWSA 7 site, ORR; centimeter scale in all photos. Sample orientations (up towards top of page) reflect tectonic dips of parent rock. **A)** Upper pit: boundary between Bw and Cr horizon showing colluvial saprolite clasts (lithorelicts) and silty soil matrix (depth —30 cm). **B)** Upper pit: intraclastic limestone saprolite with relict discoidal intraclasts (ic) (flat pebble conglomerate) inherited from parent rock and adjacent interparticle pores filled with pedogenic clays (arrows) (depth —100 cm). **C)** Upper pit: siltstone and shale saprolite with vertical bedding; note distinctive bedding-parallel (bp) and bedding-perpendicular (o) (orthogonal) fractures, as well as central area of laminated very fine-grained sandstone to coarse siltstone (depth —90 cm). **D)** Lower pit: saprolite showing complex interlayering of both sandstone (ss) and siltstone and shale (sh) saprolite lithologies. Siltstone and shale saprolite is heavily fractured and fragmented; note the large clay-filled fracture (arrow) that crosscuts the sandstone saprolite (depth —190 cm).



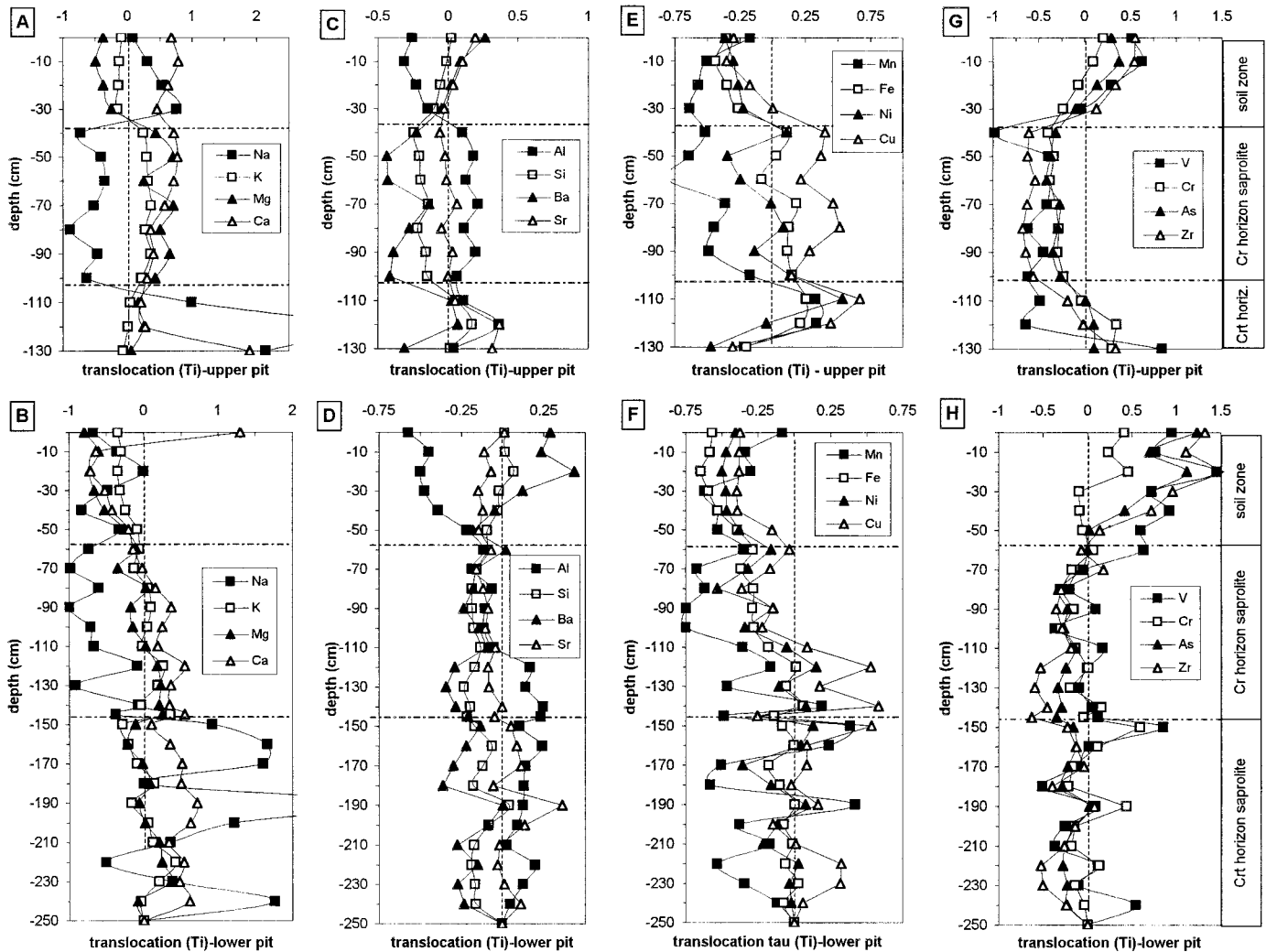


FIG. 9.—Translocation functions for soil and saprolite at SWSA 7 site, ORR, calculated assuming Ti as an immobile tracer and the deepest saprolite sampled from the lower pit at 250 cm depth as the model parent material. **A, B**) Translocations for exchangeable base cations. Note enrichments in Na in deep saprolite and inverse relationship with K, which define position of zone of pedogenic clay (disordered, randomly interlayered smectite–chlorite or smectite–vermiculite) accumulation. **C, D**) Translocations for Al, Si, Ba, and Sr showing increases in Si and Ba in solum due to residual enrichment and leaching, respectively, and increases in Al in deep saprolite due to clay translocation. **E, F**) Translocation functions for redox-sensitive elements (Mn, Fe, Ni, Cu), which are enriched in shallow and deep saprolite (upper pit) and deep saprolite (lower pit) due to fluctuating Eh caused by perched water table formation during intense rainfall events. **G, H**) Translocation functions for biocycled trace elements (V, Cr, As), which are enriched in the solum by biocycling and in the deep saprolite, to a lesser degree, by variable redox potential; net gains of Zr in solum are due to residual enrichment and concentration of detrital zircon grains.

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FIG. 8.—Photomicrographs of bedrock, soil, and saprolite at SWSA 7 site, ORR. **A**) Lower pit Bw horizon showing subangular colluvial saprolite fragments (fr) (lithorelicts) derived from upslope Cr horizon and embedded in soil matrix that consists largely of very fine sand and silt-size quartz and opaque Fe/Mn-oxides (depth –35 cm). Plane-polarized light (PPL). **B**) Lower pit Bw horizon showing pedogenic clay coating (arrow) reworked into loamy soil matrix containing subangular saprolite fragments (fr) (lithorelicts) (depth –35 cm). (PPL). **C, D**) Upper pit sandstone and siltstone and shale saprolite (Cr horizon) showing bedding-parallel (bp) and bedding-perpendicular (o) (orthogonal) fractures partially filled with pedogenic clay (depth –95 cm). (PPL and cross-polarized light (XPL), respectively). **E**) Upper pit sandstone saprolite (Cr horizon) with root pore (rp) almost completely filled with pedogenic clay coating (cl) showing meniscate (“pendant”) morphology, opaque Fe/Mn-oxide cement, and altered glauconite grains (g) (depth –100 cm). (PPL). **F**) Upper pit siltstone and shale saprolite (Cr horizon) showing bedding-parallel fracture filled with birefringent compound clay (cl) and silt pore filling (coating) (depth –110 cm). (XPL). **G**) Upper pit sandstone saprolite (Cr horizon) with bedding-parallel fracture pore infilled with pedogenic clay (cl) and showing partially weathered glauconite grains (g), which are probable precursor for at least some pedogenic clays (depth –110 cm). (PPL). **H**) Lower pit, parent material bedrock (calcareous sandstone to sandy peloidal limestone, R horizon) with calcite spar cement-filled fracture (cc), 100 μ m wide, that crosscuts bedding; note abundant quartz grains and calcite spar cement (depth –350 cm). (XPL).

clay-rich limestone saprolite (in which very few fractures were visible) the fluorescence occurred in scattered clusters and isolated spots of high intensity. This suggests that there are macropores in both lithologies but that they are very different in distribution, morphology, and, probably, origin.

We hypothesize that the large variability in measured values of bulk hydraulic conductivity, nearly four orders of magnitude (Fig. 2), is chiefly caused by three basic factors: (1) density and interconnection of fractures, root holes, and other macropores, (2) degree of development of pedogenic clay and redoximorphic (Fe/Mn oxide) features, and (3) the relative proportions of the three primary saprolite lithologies (i.e., sandstone, siltstone and shale, limestone). All of these factors should be considered when evaluating the hydrology of saprolite because, as shown in this study, fracture or macropore data alone can be a poor indicator of high or low hydraulic conductivity materials. Preliminary examinations of thin sections prepared from experimental columns with measured low hydraulic conductivities tentatively support these hypotheses (McKay and Driese 1999). Recent *in situ* measurements of both K_{sat} and unsaturated hydraulic conductivity ($K\psi$) at various water tensions confirm that clay infillings reduce conductivity by at least two orders of magnitude in an interbedded limestone and shale saprolite (Smith et al. 2000).

Geochemistry and Hydrology

The geochemical mass-balance data also suggest a three-zone (soil zone, shallow saprolite, deep saprolite) weathering system that agrees with the Solomon et al. (1992) flow model (Figs. 1, 9). On the basis of element translocation patterns we infer that the upper two zones represent a region of maximum flow and that the lower zone (saprolite below 1–1.5 m depth) represents a region of lower hydraulic conductivity and reduced flow (Fig. 9). This interpretation is based on large positive translocations for pedogenic clay components and redox-sensitive trace elements in the deep saprolite zone (Crt horizon), which suggest that major changes in lithology or reduced porosity and permeability occur in these regions. Furthermore, the inflections suggest that the deep saprolite zone contains clay accumulations greater than the amount concentrated in the shallow zone (Cr horizon). Specific evidence suggesting a clay-rich deep saprolite zone is abundant. Leaching is reduced to its lowest values in the deep saprolite zone, judging by Ba and Sr translocations. Na and K, each of which commonly fills the cation site in clay lattices, have significant translocations; although depletions in these elements occur in the shallow saprolite zone, they are highly enriched in the deep zone. Similar net enrichments in redox-sensitive trace elements, which are adsorbed on clays, accompany the changes in Na and K.

Both the geochemical and the petrographic data therefore reinforce interpretations about clay accumulation in the deep saprolite zone (Crt horizon). An additional complication, however, is our preliminary identification of the pedogenic clay minerals as a highly disordered, randomly interlayered, smectite–chlorite or smectite–vermiculite mixture (O.C. Kopp, personal communication 1999), which was confirmed with XRD analysis. The geochemical data revealed an inverse relationship between Na and K, an exchange common with smectites (Borchardt 1989). This disordered clay with smectite interlayers is interpreted as having precipitated in the pedogenic environment from weathered glauconite, because glauconite can transform directly to smectite without first existing as vermiculite (Robert 1972). Although Kooner et al. (1995) reported a disordered smectite present in amounts up to 37 wt % in a ridgetop clay lens at the ORR, most other reports neither mentioned nor document the presence of any smectites (Rothschild et al. 1984; Phillips et al. 1998). Clay with smectite interlayers is important to the flow regime, especially if alternating periods of saturation and aeration (dehydration) occur. Aeration can cause the clay lattice to contract, which may lead to cracking and increased porosity. Subsequent re-wetting of the clay in a fracture may cause swelling significant enough to plug pores, which could decrease porosity.

CONCLUSIONS AND IMPLICATIONS

Our study clearly indicates that clogging of fractures, macropores, and matrix pores with pedogenic clays and Fe/Mn-oxides has occurred in saprolite formed from sedimentary rock and that the clay and Fe/Mn-oxides play a major role in controlling hydraulic conductivity and the formation of the stormflow zone at the SWSA 7 site at the ORR. Variability in the degree of infilling contributes substantially to the wide variation in measured bulk hydraulic conductivity values measured in the field, as well as in experimental columns of fractured sedimentary saprolite sampled from the same sites (Fig. 2; Table 2). Additionally, contrasting petrographic features such as the distribution and morphology of clay within matrix and fracture pores, and the varying redox environments at different locations attested to by Fe/Mn-oxide coatings and pore fillings, provide insight concerning the contrasts in geochemistry of materials and flow regime that are unique to saprolite depth (Figs. 6–9). Specifically, net gains in Na, K, and Al at a saprolite depth of 1–1.5 m correlate with increased clay pore fillings; similarly, net gains in Fe, Mn and other redox-sensitive trace elements correlate with increased Fe/Mn pore fillings. Petrography and mass-balance chemical reconstruction are both useful for developing an understanding about complex hydrogeologic systems. Agreement between our interpretations and those of Solomon et al. (Fig. 1; 1992) and Luxmoore et al. (1990) further suggests that the combined petrographic and geochemical mass-balance methods presented here are viable and useful in studies pertaining to the hydrogeology of fractured sedimentary saprolite, especially in evaluating relative importance of macropore vs. matrix flow and formation of zones of low hydraulic conductivity in the shallow vadose zone.

Shallow subsurface flow at the SWSA 7 site therefore appears to be compartmentalized into at least three zones, consisting of 20–60 cm thick soil zone (A and Bw horizons), a shallow saprolite (Cr horizon), and deep saprolite (Crt horizon; >100 to 150 cm depth). Clay translocation and precipitation in the fracture and matrix pores contribute to significant porosity reduction and impedance of vertical drainage. Drainage impedance causes temporary water-table perching and variable redox conditions. It is likely that there is additional petrologic, geochemical, and hydrologic zonation in the saprolite below the maximum depth of this study (2.5 m), particularly near the saprolite–bedrock transition, but this has not yet been documented.

The environmental implications of this study are quite broad. The presence of a shallow zone of pore occlusion in saprolite derived from all three parent materials (limestone, siltstone and shale, and sandstone) suggests that this may be a common characteristic, although perhaps to varying degrees, of highly weathered sedimentary rocks in humid, temperate environments. The zone of occlusion is analogous in many respects to the zone of extensive calcite deposition (K horizon, calcrete, or caliche) found in many arid soil environments (Goudie 1983; Machette 1985). This suggests that conceptual models developed for evaluating runoff and infiltration in caliche-dominated settings may also be relevant in the humid southeastern U.S. In particular, we expect that in areas where the zone of occlusion is well developed there will be less uniform infiltration, more rapid surface runoff, and a tendency for shallow subsurface flow to converge in low-lying areas such as sinkholes or swallowholes. This is likely a contributing factor in the low bedrock recharge rates (Solomon et al. 1992) and frequent occurrence of ephemeral springs in the Oak Ridge area (Jones 1995).

As previously mentioned, rapid flow in the stormflow zone at ORR has been implicated in lateral spreading of waste from numerous shallow disposal trenches (Luxmoore and Ferrand 1993). It is likely that this also causes rapid downslope migration of contaminants from many other shallow sources such as septic fields, agricultural chemical applications, and industrial chemical spills. As a result, efforts to characterize or remediate contamination must recognize the possibility that lateral spreading of contaminants can be much greater than that typically observed in other hydro-

geologic settings. This study suggests that the potential for development of a zone of low hydraulic conductivity in the subsoil and the resulting active stormflow zone can be evaluated, at least on a preliminary basis, by observing the extent and degree of fracture and macropore infilling in samples from test pits or borings. These observations are much faster and less expensive than the seasonal water-level monitoring, tracer experiments, or *in situ* measurements of hydraulic conductivity previously used to investigate the stormflow zone at ORR (Wilson and Luxmoore 1988; Moore 1988; Wilson et al. 1990; Wilson et al. 1993; Luxmoore et al. 1990; Luxmoore and Ferrand 1993). As a result, these types of sedimentologic–pedologic investigations can be used as a survey method to delineate the soil types and areas where development of an active stormflow zone is likely to occur.

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APPENDIX 1.—Whole-rock geochemistry (wt %) and bulk density (g/cm^3) of soil and saprolite samples from SWSA 7, ORR.

Sample	Depth (cm)	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₂ O ₅	K ₂ O	CaO	TiO ₂	V	Cr	MnO	Fe ₂ O ₃	Co	Ni	Zn	Pb	As	Rb	Sr	Zr	Ba	Cu	Total	Bulk. Dens.
ORNL-SWSA 7-SUTITE A1																									
098A1	0	0.45	0.97	13.32	66.63	0.173	3.43	0.27	0.91	0.010	0.013	0.193	5.31	0.001	0.007	0.019	0.003	0.001	0.013	0.008	0.048	0.088	0.004	91.88	1.13
10A1	-10	0.58	0.83	13.12	68.81	0.171	3.52	0.31	0.97	0.011	0.013	0.122	4.95	0.001	0.008	0.018	0.002	0.001	0.013	0.008	0.051	0.081	0.004	93.61	1.44
20A1	-20	0.40	1.25	15.21	66.67	0.145	3.98	0.20	0.94	0.016	0.016	0.127	5.51	0.002	0.008	0.022	0.003	0.001	0.014	0.007	0.038	0.088	0.005	94.65	1.50
30A1	-30	0.78	1.24	16.42	62.83	0.131	3.44	0.26	0.98	0.007	0.009	0.091	6.53	0.002	0.009	0.021	0.002	0.001	0.013	0.007	0.038	0.072	0.007	92.88	1.50
40A1	-40	0.12	2.27	20.32	50.89	0.116	4.85	0.29	0.94	0.000	0.007	0.116	9.47	0.003	0.013	0.030	0.002	0.001	0.018	0.007	0.013	0.056	0.009	89.54	1.60
50A1	-50	0.25	2.68	21.68	53.21	0.106	5.01	0.30	0.94	0.004	0.008	0.085	8.72	0.003	0.008	0.019	0.001	0.001	0.018	0.007	0.012	0.041	0.009	93.10	1.85
60A1	-60	0.28	2.02	21.14	54.98	0.111	5.19	0.29	0.95	0.004	0.007	0.047	7.92	0.002	0.009	0.016	0.001	0.001	0.018	0.007	0.015	0.042	0.008	93.07	2.41
70A1	-70	0.19	2.50	20.68	53.43	0.115	4.91	0.25	0.87	0.004	0.007	0.142	9.31	0.003	0.011	0.030	0.002	0.001	0.018	0.007	0.011	0.058	0.009	92.56	2.06
80A1	-80	0.05	2.27	19.52	49.98	0.117	4.72	0.22	0.89	0.002	0.008	0.125	9.09	0.003	0.012	0.031	0.002	0.001	0.016	0.006	0.010	0.050	0.009	87.13	1.78
90A1	-90	0.22	2.48	20.89	53.54	0.115	4.98	0.22	0.89	0.003	0.008	0.116	8.98	0.003	0.010	0.027	0.002	0.001	0.018	0.007	0.011	0.042	0.008	92.58	2.22
100A1	-100	0.15	2.10	18.21	53.22	0.129	4.41	0.20	0.87	0.002	0.008	0.184	9.08	0.003	0.012	0.027	0.002	0.001	0.016	0.006	0.013	0.039	0.007	88.69	2.04
110A1	-110	0.70	1.53	16.95	59.06	0.146	3.36	0.17	0.78	0.003	0.009	0.264	8.86	0.003	0.015	0.030	0.003	0.001	0.014	0.006	0.022	0.061	0.009	92.00	1.31
115A1	-115	0.49	1.42	16.68	54.29	0.156	2.71	0.16	0.74	0.000	0.007	0.282	9.95	0.003	0.015	0.034	0.004	0.001	0.013	0.006	0.020	0.047	0.009	87.04	1.22
120A1	-120	1.41	1.46	18.38	57.61	0.135	2.87	0.16	0.69	0.002	0.011	0.235	7.55	0.002	0.008	0.020	0.003	0.001	0.013	0.007	0.023	0.056	0.007	90.63	1.29
130A1	-130	1.21	1.51	17.31	61.61	0.172	3.27	0.44	0.85	0.011	0.013	0.170	6.13	0.002	0.006	0.015	0.002	0.001	0.011	0.008	0.039	0.045	0.004	92.83	1.86
140A1	-140	2.05	0.69	13.14	73.18	0.206	2.70	0.18	0.58	0.015	0.018	0.178	4.32	0.001	0.004	0.003	0.002	0.002	0.008	0.012	0.027	0.097	0.003	97.42	1.57
ORNL-SWSA 7-SUTITE A2																									
0-98A2	0	0.16	0.86	13.16	59.95	0.176	3.30	0.73	0.96	0.012	0.013	0.266	5.48	0.001	0.008	0.017	0.003	0.001	0.013	0.010	0.042	0.097	0.005	85.26	1.13
5-98A2	-5	0.33	0.88	13.20	63.47	0.177	3.25	0.42	0.97	0.009	0.007	0.253	5.20	0.001	0.006	0.007	0.002	0.001	0.012	0.009	0.047	0.084	0.003	88.33	1.44
10-98A2	-10	0.56	0.94	14.02	67.69	0.160	3.43	0.21	0.99	0.008	0.010	0.181	5.08	0.001	0.006	0.010	0.002	0.001	0.015	0.008	0.051	0.076	0.003	93.45	1.80
20-98A2	-20	0.38	1.28	17.03	64.22	0.134	4.20	0.26	1.04	0.008	0.010	0.099	5.66	0.002	0.006	0.010	0.002	0.001	0.015	0.008	0.035	0.069	0.005	94.47	1.64
30-98A2	-30	0.33	1.48	17.51	60.76	0.128	4.13	0.29	1.06	0.006	0.009	0.047	6.64	0.002	0.008	0.020	0.002	0.001	0.016	0.007	0.027	0.047	0.006	92.51	1.62
40-98A2	-40	0.20	2.09	20.36	52.53	0.114	4.75	0.35	0.95	0.003	0.008	0.064	8.56	0.003	0.011	0.027	0.001	0.001	0.018	0.007	0.014	0.035	0.008	90.10	1.87
50-98A2	-50	0.26	2.04	18.81	55.20	0.123	4.40	0.33	0.92	0.001	0.007	0.066	8.42	0.003	0.011	0.024	0.002	0.001	0.017	0.007	0.017	0.048	0.007	90.70	1.88
60-98A2	-60	0.24	2.13	19.98	53.74	0.116	4.54	0.28	0.94	0.003	0.006	0.061	8.81	0.003	0.012	0.028	0.002	0.001	0.019	0.007	0.013	0.054	0.008	90.98	1.36
70-98A2	-70	0.15	2.32	21.35	53.81	0.112	5.05	0.27	0.96	0.000	0.005	0.059	9.77	0.003	0.014	0.030	0.002	0.001	0.019	0.006	0.009	0.046	0.010	92.99	1.70
80-98A2	-80	0.03	2.03	19.80	52.90	0.116	4.85	0.24	0.95	0.001	0.007	0.094	9.11	0.003	0.013	0.026	0.002	0.001	0.019	0.006	0.012	0.031	0.010	90.26	1.64
ORNL-SWSA 7 SUTITE B1																									
0B	0	0.15	0.35	8.33	71.91	0.266	2.69	0.41	0.99	0.013	0.017	0.233	3.88	0.001	0.007	0.022	0.003	0.002	0.010	0.007	0.079	0.098	0.004	89.50	1.19
10B	-10	0.31	0.72	11.52	77.45	0.180	3.08	0.07	1.07	0.013	0.016	0.181	4.01	0.001	0.007	0.017	0.002	0.002	0.011	0.007	0.077	0.101	0.005	98.68	1.37
20B	-20	0.48	0.49	10.30	80.13	0.200	2.85	0.05	1.05	0.018	0.019	0.188	3.36	0.001	0.007	0.013	0.002	0.002	0.011	0.007	0.091	0.116	0.004	99.39	1.53
30B	-30	0.26	0.61	11.14	75.89	0.182	3.08	0.09	1.08	0.013	0.012	0.105	3.99	0.001	0.007	0.020	0.003	0.002	0.011	0.007	0.072	0.093	0.004	96.67	1.43
40B	-40	0.08	0.83	12.43	72.50	0.164	3.28	0.11	1.04	0.014	0.012	0.125	4.44	0.001	0.007	0.019	0.002	0.001	0.012	0.007	0.061	0.076	0.004	94.47	1.52
50B	-50	0.32	1.33	16.02	67.70	0.134	3.95	0.15	1.04	0.012	0.012	0.125	5.72	0.002	0.008	0.018	0.002	0.001	0.015	0.007	0.041	0.066	0.006	96.69	1.54
60B	-60	0.01	1.48	16.89	62.97	0.123	3.78	0.15	0.97	0.011	0.013	0.162	6.28	0.002	0.010	0.025	0.003	0.001	0.015	0.007	0.031	0.076	0.006	93.13	1.40
70B	-70	0.01	1.16	16.75	63.49	0.122	3.79	0.19	1.05	0.007	0.011	0.087	6.00	0.002	0.009	0.021	0.002	0.001	0.015	0.007	0.042	0.067	0.006	92.82	1.55
80B	-80	0.19	1.81	19.09	60.65	0.112	4.53	0.22	1.04	0.006	0.009	0.102	6.76	0.002	0.006	0.011	0.002	0.001	0.017	0.007	0.025	0.065	0.005	94.64	1.88
90B	-90	0.00	1.40	17.45	58.18	0.118	4.52	0.25	1.00	0.008	0.010	0.064	6.44	0.002	0.011	0.023	0.002	0.001	0.017	0.007	0.022	0.059	0.006	89.57	2.02
100B	-100	0.13	1.46	17.59	59.88	0.119	4.38	0.23	1.01	0.004	0.009	0.065	6.61	0.002	0.008	0.020	0.002	0.001	0.017	0.007	0.025	0.067	0.005	91.64	1.57
110B	-110	0.14	1.66	17.79	59.38	0.119	3.89	0.20	0.96	0.008	0.010	0.158	7.14	0.002	0.011	0.026	0.002	0.001	0.015	0.007	0.027	0.067	0.007	91.63	1.55
120B	-120	0.39	1.88	21.56	55.87	0.104	4.90	0.26	0.94	0.007	0.012	0.202	8.63	0.002	0.014	0.031	0.003	0.001	0.018	0.006	0.015	0.051	0.010	94.90	1.82
130B	-130	0.04	1.95	21.15	51.96	0.103	4.63	0.23	0.94	0.006	0.009	0.130	8.08	0.002	0.011	0.026	0.003	0.001	0.018	0.006	0.013	0.048	0.008	89.37	1.47
140B	-140	0.39	1.90	22.71	53.56	0.099	3.70	0.22	0.93	0.007	0.013	0.283	8.94	0.003	0.013	0.034	0.004	0.001	0.017	0.007	0.017	0.051	0.010	92.92	1.45
140GB	-145	0.27	2.03	23.23	54.03	0.095	5.40	0.27																	